DIET

01/09/22

Balanced diet:-

- eating all the food groups in the right proportions and amount.

importance:-

carbohydrates:

* body's primary source of energy
* broken down into glucose which is used as fuel.

Fats and oils:

* provide energy
* insulate the body

proteins:

* repair and build body's tissues
* allows metabolic reactions to take place

vitamin C:

* helps protects cells and keeps them healthy.

vitamin D:

* helps the body absorb and retain calcium and phosphorus.

calcium:

* builds strong bones and teeth
* clots blood faster on injured area to prevent loss of blood

iron:

* needed for growth and development
* makes hemoglobin, myoglobin and hormones

Fibre:

* helps prevent constipation
* lowers cholesterol levels

water:

* carries nutrients and oxygen to the cell
* helps cool down the body

•- scurvy:

- the deficiency of vitamin C

* swollen, bleeding gums

•- rickets:

- the deficiency of vitamin D

* weak, soft and bendy bones

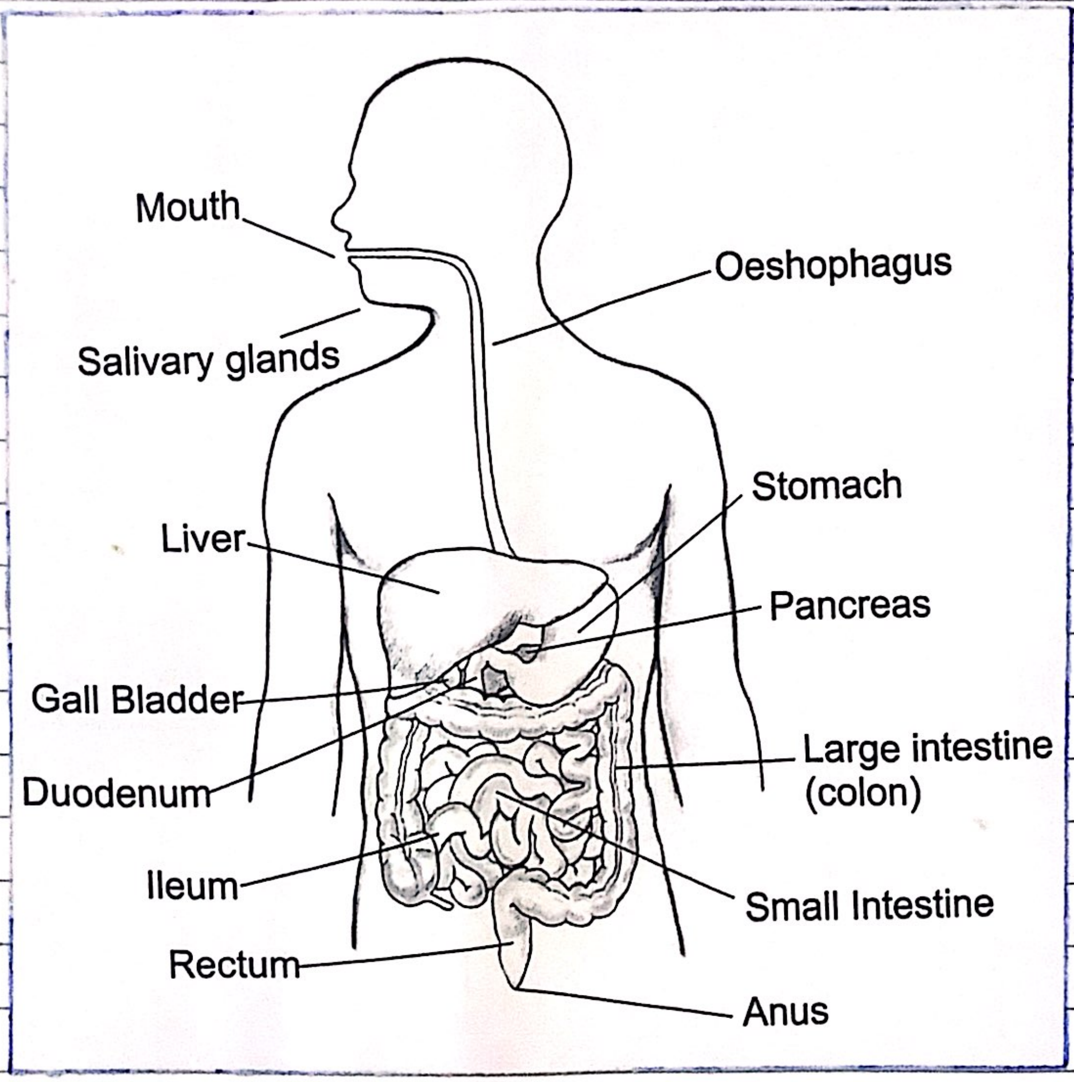
•- kwashiorkor

- the deficiency of protein

* swollen belly

DIGESTIVE SYSTEM

19/09/22



Mouth:-

Performs chemical and mechanical digestion. Mechanical by teeth grinding food. Chemical by the saliva which breaks down starch using amylase.

Salivary glands:-

stores the saliva that has amylase.

Oesophagus:-

Food is pushed through the oesophagus using peristalisis which is a series of wave-like muscle contractions that move the food.

20/09/22

Stomach:-

Produces HCL which kills bacteria and makes a suitable environment for pepsin to split up the proteins. The walls of the stomach also mash food. They contract.

Liver:-

produces bile which dissolves large insoluble fat globules into small soluble fat globules. It also processes nutrients from the blood and clean the blood from poison and waste

Gall bladder:-

stores the bile produced by the liver.

Pancreas:-

produces enzymes such as amylase, protease and lipase which the small intestine needs to further break down the food. It also makes insulin which controls the sugar level in the blood.

Small intestine:-

The start of the small intestine is the duodenum. The food enters from here. Then it is absorbed for all the nutrients and water. The ending of the small intestine is known as ileum.

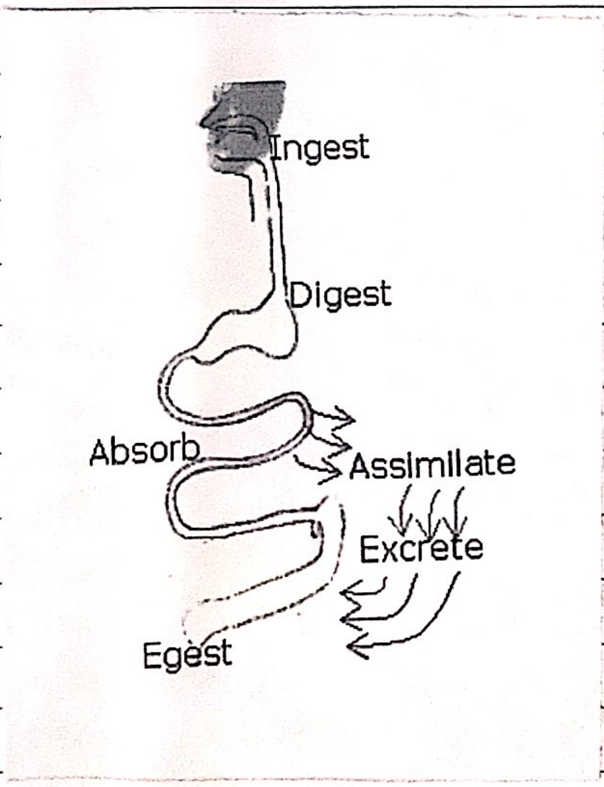
Large intestine:-

absorbs the undigested water and salts. Moves the waste material to the rectum which stores it until it is pushed out through the anus as faeces.

21/09/22

5 stages

ingestion:-
taking in of substances e.g food and drink, into the body
digestion:-
the break down of food
absorption:-
the movement of the nutrients from the food, from the intestines to the blood.
assimilation:-
uptake and use of nutrients by the cells
egestion:-
the removal of undigested food / waste from the body as faeces.



PHYSICAL DIGESTION

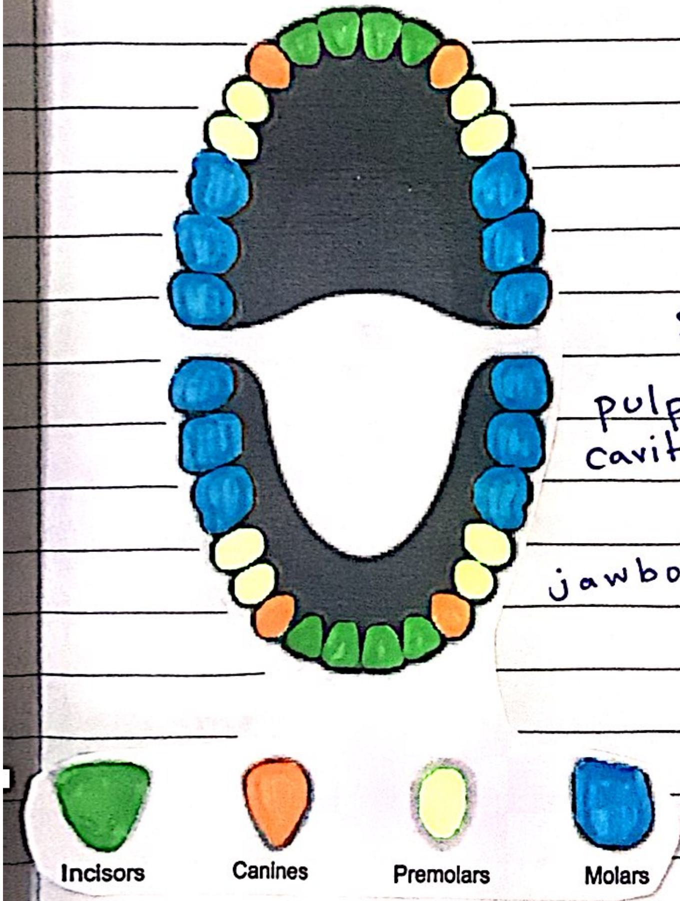
22/09/22

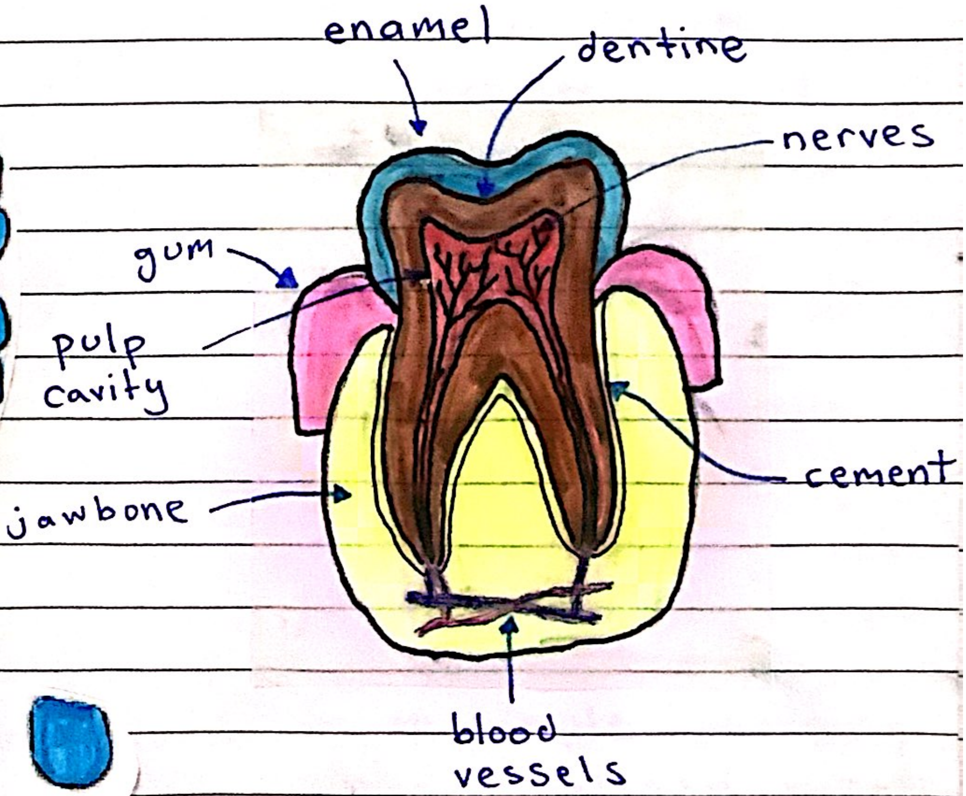
Physical digestion:-

- breakdown of food into smaller pieces without chemical change to the food molecules.

- it increases the surface area of food for the enzymes in chemical digestion

- mainly, the physical digestion is done by the teeth.





incisors :- used to bite into the food
canines :- sharp to tear apart the food
premolars :- used to tear and crush the food
molars :- used to grind, tear and crush food

07/11/22

role of stomach in physical digestion:-

- physically breaks down the food into smaller particles by using the muscle in the stomach Lining to churn and squeeze the food.

role of bile in chemical digestion:-

- bile is used to turn Large insoluble fat globules into small soluble fat globules
- fat → fatty acids
- molecules of bile are attracted to fat and water, so the bile sandwiches them together, thus fat molecules are kept seperated and lipase is able to digest them.

CHEMICAL DIGESTION

- used to break down large insoluble molecules into small soluble molecules that can be absorbed

enzymes:-

amylase = breaks down starch to simple reducing sugars such as maltose. Maltase breaks down maltose into glucose in small intestine.

protease = breaks down proteins into amino acids.

lipase = breaks down fats and oils to fatty acids and glycerol

* all 3 made in pancreas (amylase made in mouth too)

07/11/22

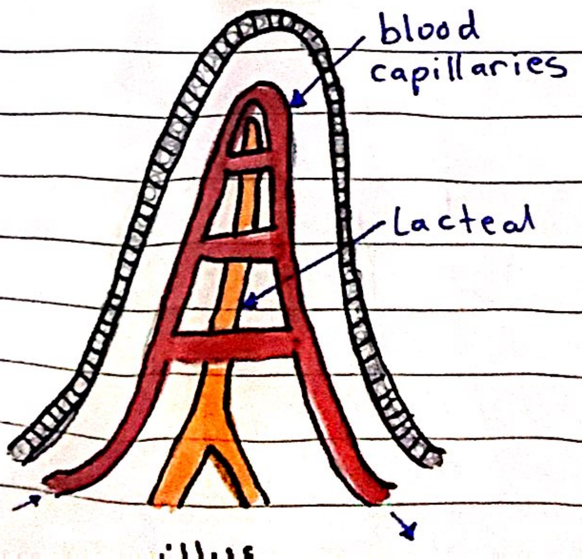
→ bile is an alkaline mixture (not enzyme) that neutralizes the acidic mixture of food and gastric juices entering the duodenum to provide a suitable pH for enzyme action.

protease =
- pepsin → breaks down protein in the acidic conditions of stomach.
- trypsin → breaks down protein in the alkaline conditions of small intestine.

hydrochloric acid (HCl):-
- kills harmful bacteria
- provides optimum pH for optimum enzyme activity

ABSORPTION

small intestine:-
- region where nutrients are absorbed into bloodstream
- most water is absorbed here (some in the colon)
- villi increase the surface area which results in more absorption.
- the lining of villi have more micro villi
- villi have a very good blood supply



* blood capillaries absorb most food molecules
* lacteal absorbs digested lipids